Comparing Unicompartmental Knee Arthroplasty and High Tibial Osteotomy for Isolated Medial Compartment Knee Osteoarthritis

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Abstract

» Both unicompartmental knee arthroplasty (UKA) and high tibial osteotomy (HTO) allow for compartment-specific intervention on an arthritic knee joint that preserves bone stock and native soft tissue compared to a total knee arthroplasty (TKA). Both operations give a more natural feeling with native proprioception compared with a TKA.

» HTO is better suited in patients who are younger (<55 years-of-age), have a body mass index (BMI) <30 kg/m^2, high activity requirements, mechanical malalignment, asymmetric varus, isolated anterior cruciate ligament insufficiency, need for multiplanar correction, and a preference for joint preserving interventions. Recent data suggest that age (>55 years-of-age) should not solely contraindicate a HTO.

» UKA may be chosen in patients who are older (>55 years-of-age), low activity requirements, have a BMI <40 kg/m^2, severe osteoarthritis with significant joint space narrowing, acceptable coronal alignment, symmetric varus, and patient preference for arthroplasty.

Unicompartmental knee arthroplasty (UKA) and high tibial osteotomy (HTO) are mainstay treatments for patients with isolated medial compartment arthrosis (Fig. 1). Both operations offer shared and unique benefits in the appropriately indicated population compared with total knee arthroplasty (TKA). Between 2007 and 2011, in the United States, UKA utilization increased by 5% annually, while HTOs declined by 4% annually[1]. There are variable patterns in HTO and UKA utilization in different countries, with Korea and Japan both having increases in annual HTO and UKA rates and a drop in TKA[2,3]. Ambulatory surgery centers, in the United States, have seen a growth in the proportion of outpatient UKA being performed between 2013 (920 UKAs) and 2018 (11,080 UKAs)[4]. Australian registry data from 2021 show that UKA procedures make up 5.7% of all knee arthroplasties, down from a peak of 14.5% in 2003[5]. There is a paucity of data regarding more current utilization rates of HTO within the United States.

The technology, techniques, and implants used in both HTO (Fig. 2) and UKA (Fig. 3) have improved since each surgical procedure's advent. Within the realm of UKA, implant design, computer navigation, and surgical technique have allowed for an expansion of indicated patients, which was historically more restrictive based on failure rates of early implants[6]. Similarly, the increased availability of locking titanium plates and bone graft material and adjuncts have improved structural integrity of HTOs[7,8].

There is substantial, and increasing, overlap in the patient populations that may

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Fig. 1
Anterior-to-posterior knee radiograph of isolated medial compartment knee arthritis.

benefit from either an HTO or UKA. There is currently a lack of randomized comparative studies to inform recommendations, and a substantial amount of the studies reviewed below are level 4 studies and large national registry databases. This article analyzes recent studies with a focus on patient characteristics and surgeon factors that affect the outcome of each procedure to help delineation of optimal treatment.

High Tibial Osteotomy

Patients with isolated medial compartment arthrosis can benefit from valgus-producing osteotomy either in the form of medial opening wedge HTO or lateral closing wedge HTO. Medial opening wedge HTO provides more precise angular correction of deformity while lateral closing wedge HTO confers greater construct stability with direct bone apposition. Traditionally, patients who are indicated for HTO are chronologically and physiologically young, active patients with no ligamentous compromise and a normal, pain-free range of motion of the knee with unicompartmental arthritis. The indications for HTO have broadened with improved techniques and implants, mainly increased biomechanical properties of modern locking plate technology[7]. Traditional contraindications for HTO are obesity, knee stiffness, ligamentous instability that is not amenable to reconstruction, inflammatory arthritis, medial bone loss, symptomatic arthritis in other compartments, or prior lateral meniscectomy[9]. Coexisting ligamentous or soft tissue pathology such as anterior cruciate ligament (ACL) or posterior cruciate ligament (PCL) deficiency, chondral pathology, or medial meniscal pathology is not necessarily contraindicated because reconstruction, repair, or replacement can be performed concomitantly or in a staged fashion[10,11], as adjustments in coronal and sagittal alignment can be used to provide a biomechanical advantage.

Unicompartmental Knee Arthroplasty

Traditional indications for UKA, originally described by Kozinn and Scott in 1989, were largely influenced by high rates of early-generation UKA failures. Early indications were patients younger than 60 years, no heavy labor, weight under 180 pounds, without heavy labor requirements, and range of motion of at least 90° with a maximum of 5° flexion contracture and coronal deformity less

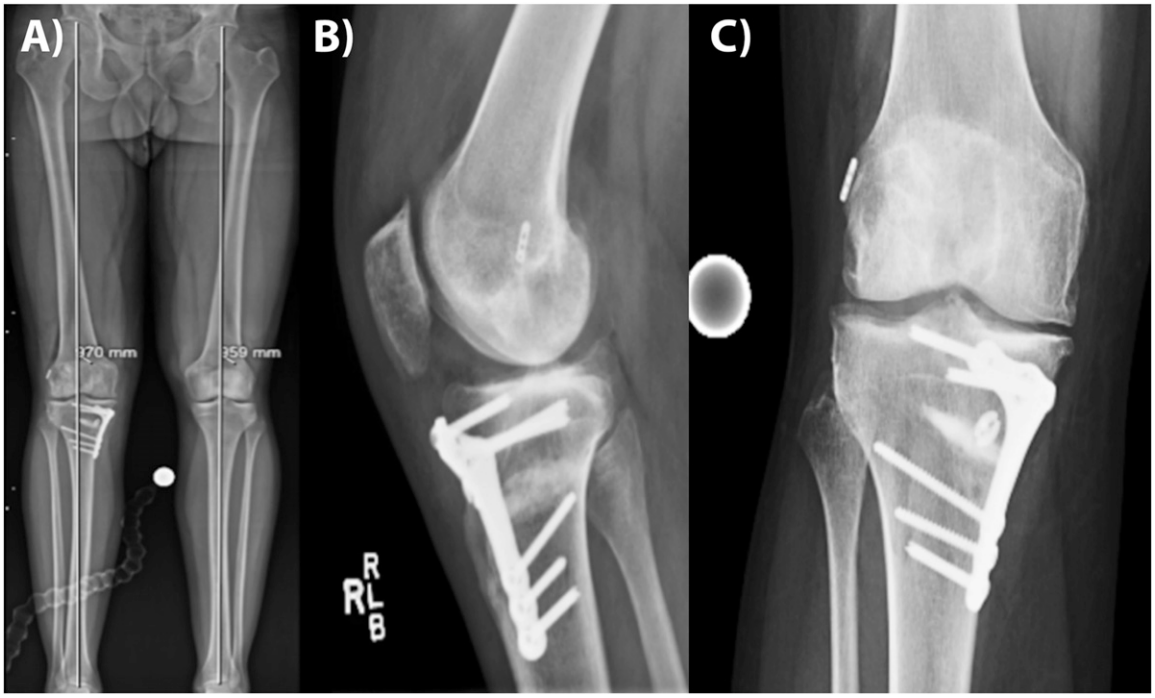

Fig. 2
Fig. 2-A Hip-to-ankle standing radiograph of a medial opening wedge high tibial osteotomy with modern locking plate technology. **Fig. 2-B** Lateral radiograph of a medial opening wedge high tibial osteotomy with modern locking plate technology. **Fig. 2-C** Anterior-to-posterior radiograph of a medial opening wedge high tibial osteotomy with modern locking plate technology.

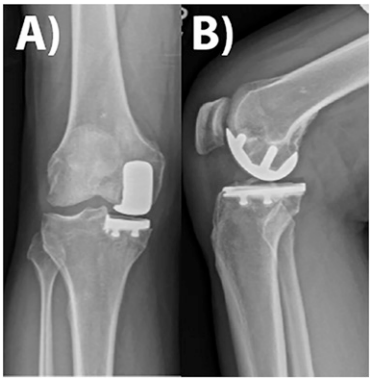

Fig. 3

Fig. 3-A Anterior-to-posterior radiograph of medial unicompartmental knee arthroplasty. **Fig. 3-B** Lateral radiograph of medial unicompartmental knee arthroplasty.

than 15°[6,12] (Table I). The corresponding contraindications from this era included presence of subchondral bone in an adjacent compartment, inflammatory arthritis, and ACL or PCL insufficiency.

A recent large prospective case series of 1,000 UKAs, performed by Hamilton et al., has challenged Kozinn and Scott's original indications[13]. They found no difference in postoperative knee scores at 10 years between patients who would be traditionally indicated and those with contraindications based on the Kozinn and Scott criteria[13]. At 10 years postoperatively, poor outcomes were statistically more frequent in so-called "indicated" knees and implant survival at 15 years was no different between groups.

Age

Traditionally, HTO is recommended in younger patients when compared with UKA[14]. Studies found improved outcomes in patients younger than 50 years with an HTO[15,16]. Trieb et al. found increasing risk of HTO failure after 65 years, with additive risk per each year of life[17]. Similarly, Pannell et al. report that patients who went on to TKA were older at the time of index HTO compared with their counterparts who had not progressed to TKA[18]. However, these studies used a combination of closing wedge osteotomies and older implant technology. Biomechanical studies have shown increase torsional and compressive stability of modern titanium locking plates compared with older, small spacer plates[7] (Fig. 4).

Recent studies, using opening wedge techniques with newer implants have not shown the same delineation in outcomes based on age. Goshima et al. prospectively found no difference in patients older or younger than 65 years regarding outcomes or postoperative course[19]. These results were echoed in a retrospective cohort study which found no difference in pain relief, satisfaction, or 15-year implant survival based on age at surgery[20].

Younger patients were not traditionally offered UKAs because of concern for implant longevity and activity demands leading to earlier revisions or poor outcomes. In a large prospective study, Hamilton et al. found patients younger than 60 years had significantly better outcomes in multiple validated knee function scores such as the American Knee Society Functional Score ($p < 0.001$), Oxford Knee Score ($p = 0.01$), and Tegner Activity Score ($p < 0.001$)[13]. In addition, there was no difference in 15-year implant survival, time to failure, or mechanism of implant failure between the under 60 and over 60-year-old cohorts. Smaller studies based on consecutive case series have found no difference in patients older or younger than 55 years regarding outcome or revision risk[21]. Contrary to these studies, 2 large meta-analyses found patients younger than 65 years had a higher risk of revision, when compared with age-matched patients undergoing primary TKA[22,23]. Recent large registry data from Australia and the United Kingdom show an increase in revision rates for patients undergoing UKA who are younger than 55 years compared with those older than 65 years[5,24].

Patient age is an important factor in the decision for either HTO or UKA. Patients younger than 50 years have consistently shown benefit after HTO, and there are mixed data which suggest that in patients older than 65 years have inferior outcomes. While advanced age is not a sole contraindication to HTO, all other patient factors should be amenable to HTO if proceeding in an older patient. UKA data from larger, modern national registries suggest that patients younger than 55 years are at substantially increased risk of revision compared with those older than 65 years at nearly all time points. UKA should be avoided in those younger than 55 years if alternative procedures, such as HTO, can be performed.

TABLE I Kozinn and Scott (1989) Original UKA "Ideal" Patient*

Indications	Contraindications
Younger than 60 yrs	Exposed subchondral bone in PFJ or opposite compartment
Low demand activity	Chondrocalcinosis
<180 lbs or 82 kg	Significant at-rest knee pain
Knee flexion ≥90° total arc with flexion contracture <5°	Inflammatory arthritis
Coronal deformity of <10° varus or <15° valgus	ACL or PCL deficiency

*ACL = anterior cruciate ligament, PCL = posterior cruciate ligament, PFJ = patellofemoral joint, and UKA = unicompartmental knee arthroplasty.

Body Habitus

Body habitus plays an important role in determining best treatment for patients with medial compartment arthritis. Obesity (body mass index, BMI >30 kg/m^2) was found in a retrospective cohort of opening wedge HTO to be associated with decrease knee function

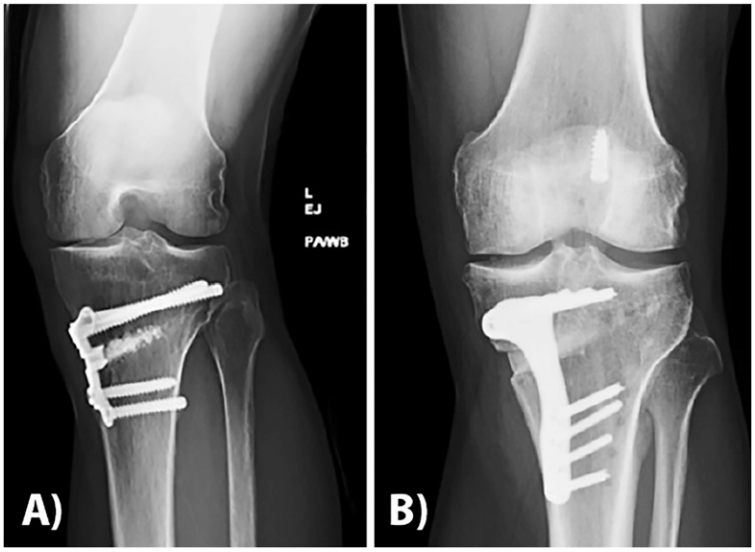

Fig. 4

Fig. 4-A Anterior-to-posterior radiograph of an older generation metal Puddu plate for fixation of a medial opening wedge high tibial osteotomy. **Fig. 4-B** Anteroposterior radiograph of a medial opening wedge high tibial osteotomy with modern locking plate technology.

scores but no increase in postoperative complications[25]. Patient obesity has also been indicated as a poor prognostic factor for postoperative knee function[26]. Lateral closing wedge HTO may be of benefit in patients with greater BMI as it allows for direct bony apposition and greater construct stability. Unfortunately, limited data exist comparing outcomes of medial opening wedge HTO with lateral closing wedge HTO in patients with elevated BMI. Efe et al. report that BMI had no correlation with conversion to TKA in patients who previously underwent lateral closing wedge HTO[27]. There are currently limited, prospective, high-quality data on the impact of BMI and outcomes of HTO, but current evidence suggests caution and thorough patient education when considering performing a HTO in a patient with obesity.

The original Kozinn and Scott weight limit of 180 pounds (82 kg) was largely based on high rates of early UKA implant failure and fear of implant stress[6]. Hamilton et al. found no difference in outcomes or implant survival between patients below or above the 180 pounds, and 2 other meta-analyses showed no increase of risk of revision with obesity (BMI >30 kg/m^2), with follow-up ranging from 2 to 18 years[13,22,28]. UKAs performed on patients with a BMI over 40 kg/m^2 showed similar rates of revision compared patients with a BMI 30 to 40 kg/m^2 at 7 years of follow-up[29]. Conversely, a recent study found a 5 times greater likelihood of major revision within 2 years in patients with BMI over 40 kg/m^2 principally related to mobile-bearing instability or progression of arthritis in the lateral compartment[30]. Taken as whole, there are certainly data to expand acceptable patient weight past the traditional 180 pounds; however, caution is warranted for patients with a BMI over 40 kg/m^2.

Patellofemoral Chondrosis

The alteration of mechanical alignment and joint contact pressures with HTO has raised concern about the progression, or development, of patellofemoral arthritis after HTO[25,26]. Studies incorporating second-look arthroscopy after opening wedge HTO showed subsequent degeneration of patellofemoral joint in up to half of patients[31-34]. This degeneration was more likely to occur in patients with large corrections (>10°) or who had an overcorrection causing valgus alignment. Goshima et al. found worsening patellofemoral arthritis on second-look arthroscopy and failed to find any clinical impact of this degeneration on patient outcomes at roughly 5 years of follow-up[31]. Kim et al. compared radiographic and clinical outcomes in patients who underwent medial opening wedge HTO with patellofemoral chondrosis and without patellofemoral chondrosis. At short-term follow-up, they did not note any significant difference in outcomes measures between the groups with similar pain and function scores[35].

An original contraindication to UKA, as purposed by Kozinn and Scott, was the presence of exposed subchondral bone at patellofemoral joint[6]. The concern was that this site of degeneration could cause continued pain after UKA and herald further arthritis progression from malalignment, which may benefit from a TKA[6]. An early study assessing 15-year follow-up on UKA found patellofemoral arthritis progression to be the primary cause of failure[36]. van der List et al. in a large meta-analysis found no difference in patients having postoperative patellofemoral arthritis and inferior outcomes or risk of UKA revision[22]. Multiple studies have found patients with pre-existing medial facet patellofemoral arthritis have no worse outcomes or revision rates after UKA[37-40]. Lateral facet arthritis of the patella should still be approached with caution when considering a UKA because postoperative alignment

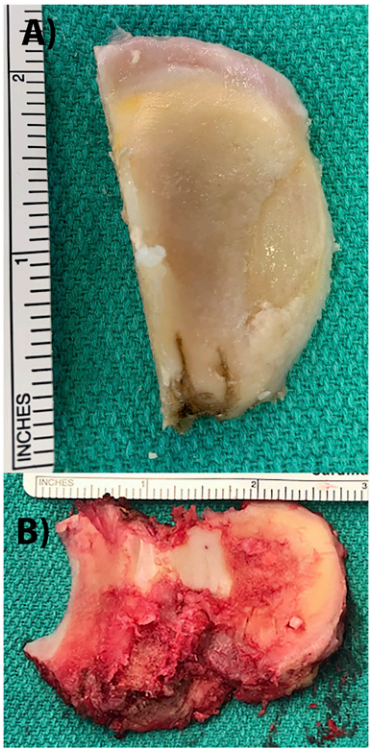

Fig. 5

Fig. 5-A Primary osteoarthritis with anteromedial wear pattern. **Fig. 5-B** Posteromedial arthritic changes associated with osteoarthritis secondary to anterior cruciate ligament injury.

changes can lead to increased contact pressures on the lateral facet.

Anterior Cruciate Ligament Integrity

ACL injury alters force distribution of the native knee and results in an increased risk of developing radiographic arthritis by 10 years compared with an uninjured knee[41,42]. Arthritis secondary to an ACL deficiency has a different natural history when compared with primary arthritis with an intact ACL. Arthritic changes secondary to ACL deficiency typically develops a posteromedial wear pattern compared with the anteromedial wear of primary arthritis[43] (Fig. 5). ACL deficiency has historically been considered a contraindication to UKA because of early aseptic loosening tibial component and progression of posteromedial arthritis leading to a fixed varus deformity and lateral compartment progression[44,45]. Finite element studies of ACL-deficient knees with a UKA show increase contact pressures on the medial UKA bearing surface and lateral compartment[46].

The clinical impact of ACL deficiency with UKA outcomes is less clear. A small, underpowered study of 46 ACL-deficient UKAs over 5 years found no difference in implant survival compared with a cohort-matched group with intact ACLs[47]. There are heterogenous data with only midterm follow-up from case series with ACL-deficient knees undergoing UKA without ACL reconstruction, which generally shows improvement in postoperative function and equivalent survival compared with UKA with ACL reconstruction[47-49]. A recent matched pair analysis with 9-year follow-up showed equal patient-reported outcomes between ACL-intact and ACL-deficient patients with UKA[50]. There is a lack of long-term follow-up on ACL-deficient patients. ACL-deficient patients should be approached cautiously given the lack of long-term data, with attention to tibial slope, subluxation, and arthritis patterns related to ACL deficiency.

In patients undergoing a HTO with concomitant ACL deficiency there is limited data on outcomes. A case series by Kim et al. assessed preoperative ACL status by MRI and first-look arthroscopy in patients undergoing HTO. Ultimately, they reported no difference in progression of arthritis in other compartments and no difference in functional knee scores based on ACL status (e.g., intact, mucoid degeneration, chronic partial tear, and complete tear) at 5 years[51].

When HTO and ACL reconstruction are performed, they can either be performed during a single surgery or as staged procedures. This delineation is made based on patient symptoms of pain and/or instability, with instability suggestive of needing to address the ACL deficiency. If the primary complaint is pain, then an HTO can be considered in isolation with subsequent ACL reconstruction if instability is later encountered. There are numerous studies showing good outcomes for combination HTO and ACL reconstruction for ACL deficiency with varus malalignment[52-56].

Posterior tibial slope has become an increasingly investigated topic regarding HTO and ACL status, with increasing posterior tibial slope (>12°, Fig. 6) being associated with increased anterior tibial translation and subsequent stress, as well as failure of ACL reconstructions[57-60]. In patients with high posterior tibial slope preoperatively, when ACL reconstruction is planned, there should be strong consideration for a decrease of the slope to protect ACL reconstruction. The use of biplanar osteotomy to correct deformity in the coronal and sagittal plane in patients undergoing concomitant ACL reconstruction is promising. Richter et al. reported improved functional outcomes up to 5 years in patients who underwent combined biplanar HTO and ACL reconstruction. Biplanar correction should be considered in patients with increased slope, medial compartment arthritis, and ACL deficiency[61].

Surgeon Factors

The risk of revision for a UKA is directly proportional to surgeon experience and caseload characteristics. In a large meta-analysis of 45 studies and 12,520 UKAs, the authors found increasing revision risk with decreasing caseload (UKA per year) and usage (% UKA of total primary arthroplasty cases)[62]. They found that usage was a more important characteristic than caseload and stated that surgeons with >30% UKA case volume would likely achieve the documented 10-year survival rate of 94%[62]. An assessment of UKA revision rates as a factor of hospital volume found that high-volume centers had UKA revision rates similar to TKA revision rates; however, low-volume centers had significantly higher UKA revision rates compared with TKA revision rates (subhazard ratio, 1.95)[63]. In a smaller study, a low-volume surgeon (<10 UKA per year, usage 34%) was more likely to have lower functional outcomes

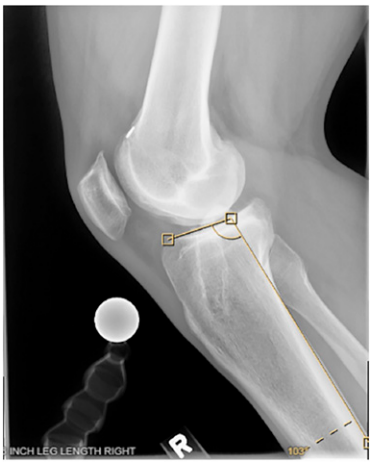

Fig. 6

Lateral radiograph of the knee demonstrating technique for measurement of posterior tibial slope. A line is drawn tangential to the posterior tibial cortex. Another line is drawn tangential to the surface of the tibial plateau. The angle created by these 2 lines equals the posterior tibial slope.

compared with a high volume (>100 UKA per year, usage 40%)[64].

There is limited information specifically related to surgeon factors, such as case volume or years of practice, and outcomes of HTOs. A large Canadian database study assessed annual HTO volume, years in practice, and hospital academic status regarding the 10-year risk of revision from HTO to TKA and found no associations after 10 years of follow-up[65].

Survivorship

Survivorship rates of opening wedge HTOs are 95% at 5-year and 92% at 10-year follow-up[66]. These results have been echoed in numerous studies with survival rates ranging from 75% to 92% at 10 years when including a combination of opening and closing wedge HTOs[67-73]. HTO survivorship may be overestimated due to the lack of large registry data seen in arthroplasty cases, which may miss the accounting of HTO to TKA conversions. Common failure risks included increasing age, female sex, and postoperative knee flexion deficits[67-70]. Osteotomy fixation techniques have continued to evolve and demonstrate good promise regarding survivorship and clinical outcomes. Keyt et al. showed survivorship free of hardware removal of 94% and survivorship free of conversion to TKA of 93% at 8 years when using modern polyetheretherketone implants[74]. Han et al. compared modern locking plate technology to nonlocking plates and showed that use of locking plates resulted in less loss of deformity correction and improved patient-reported outcomes[75].

When an HTO fails, or arthritis advances, a TKA is often the next intervention. The survivorship of a TKA after HTO is not significantly different from primary TKA, with 10-year survival rates above 90%[76,77]. Notwithstanding, the conversion from HTO to TKA is more technically challenging due to skin and soft tissue scarring, ligament imbalance, anatomical distortion, and the need for hardware removal and replacement[78]. By contrast, the revision rate of HTO to TKA conversions has been found to be higher than revision rates of primary TKAs in some meta-analyses[79,80].

UKA 10-year survivorship ranges from 87% to 94% with higher rates of survival found when causes of revision are isolated to implant-related causes of failure, as opposed to infection[81-84]. Lateral compartment arthritis progression is the most often indication for conversion to TKA, followed by aseptic loosening of the tibial component[73-76]. Lateral compartment progression is hastened with valgus overcorrection of the mechanical alignment[82]. Smaller

TABLE II Summary of Recommendations*

	HTO	UKA
Age	Younger†	Older†
Body mass index, (kg/m^2)	≤30	≤40
Activity level	High activity requirement	Low to moderate activity requirement
Joint space narrowing	Little to none	High Kellgren and Lawrence OA grade
Instability	Must exercise caution when operating on a clinically unstable joint Would defer decision to alternative factors	
Malalignment	No contraindication based on axis deviation Asymmetric varus deformation	Contraindicated in higher degree of axis deviation
Patellofemoral chondrosis	The presence or absence of patellofemoral chondrosis does not significantly affect outcomes Would defer decision to alternative factors	
ACL integrity	Reasonable with impaired and intact ACL Would defer decision to alternative factors	
Surgeon factors	Complex surgery with limited data on surgeon volume and outcomes	Decreased revision rates with surgeon/center experiencing higher case volumes
Survivorship	75%-92% at 10 yrs	87%-94% at 10 yrs

*ACL = anterior cruciate ligament, BMI = body mass index, HTO = high tibial osteotomy, OA = osteoarthritis, and UKA = unicompartmental knee arthroplasty. †HTO is classically believed to be more efficacious in younger patients and UKA in older patients. However, the most recent studies have proven that HTO can be successful in older patients and UKA in younger patients.

mobile bearings and slight under correction of mechanical alignment decreases the risk of lateral compartment overload and subsequent arthritic progression[82,85,86].

When a UKA is revised to a TKA, there are mixed data on outcomes compared with primary TKA. Two recent matched-cohort studies comparing primary TKA with TKA after failed UKA showed no difference in knee function score or implant survival at 2 and 8 years of follow-up[87,88]. A large registry study found that UKA revised to TKA had functional outcomes closer to revision TKAs, not primary TKAs[89]. In general, the technical conversion of a failed UKA to a TKA is more likely to include revision components such as stems and augments, compared with a standard primary TKA[90].

Return to Activity

Patients with medial compartment arthritis can often be quite debilitated, leading to inability to participate in work duties or time away from sport and hobbies. Agarwalla et al. report a 95% return to work rate at a mean of 3 months after surgery. They noted that patients with jobs requiring heavy duty required more time away from work compared with their counterparts with sedentary, light, and moderate duties[91]. Liu et al. reported an 88% return to sport rate after HTO; however, only 41% returned to a presurgical level of participation. Patients should be counseled that a return to lower-impact activity may be necessary after HTO[92].

UKA has been shown to have a similar return to work rate, with studies recording return to work rate between 75% and 100%, with most showing rates closer to 90% to 95%[93-96]. However, the number of sport and recreational disciplines patients were engaged in decreased. In addition, the frequency of activity has been shown to remain constant postoperatively while the duration of activity varied[93,95]. The activities most likely to be discontinued were high-impact activities, while low-impact or mid-impact activities were continued. In cases where high-impact activities must be resumed, patients should be warned about the risks of implant loosening and wearing[95].

Summary

Factors that warrant selection of HTO include younger age; high activity requirements regardless of severity of arthritis; minimal joint space narrowing with or without concomitant cartilage, meniscus, or ligament surgery; poor alignment; asymmetric varus; isolated ACL insufficiency; benefits to correcting 2 planes (e.g., change in slope after previous ACL/PCL repair); and patient preference (Table II). UKA may be chosen in patients with older age, low activity demands, severe osteoarthritis with substantial joint space narrowing, acceptable alignment, lack of asymmetric varus, and patient preference (Table II).

Conclusions

Both UKA and HTO allow for compartment-specific intervention to the arthritic knee joint. This preserves bone stock and native soft tissue compared with a TKA. Both operations give a more natural feeling with more native proprioception compared with a TKA. Delineating between either UKA or HTO remains difficult, primarily because of expanding indications for both procedures. The overlap in patient population that is eligible for either UKA or HTO allows for opportunity for shared decision-making. Patient decision aids are used to help inform patients of multiple interventions for a condition, with the aim of educating patients and helping form a surgical preference without worsening decisional conflict. These decision aids have been used with success in smaller studies of medial knee OA with treatment options including UKA or TKA[97]. The development of a decision aid for UKA or HTO is an area of potential future work given the overlap in patient population and outcomes.

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